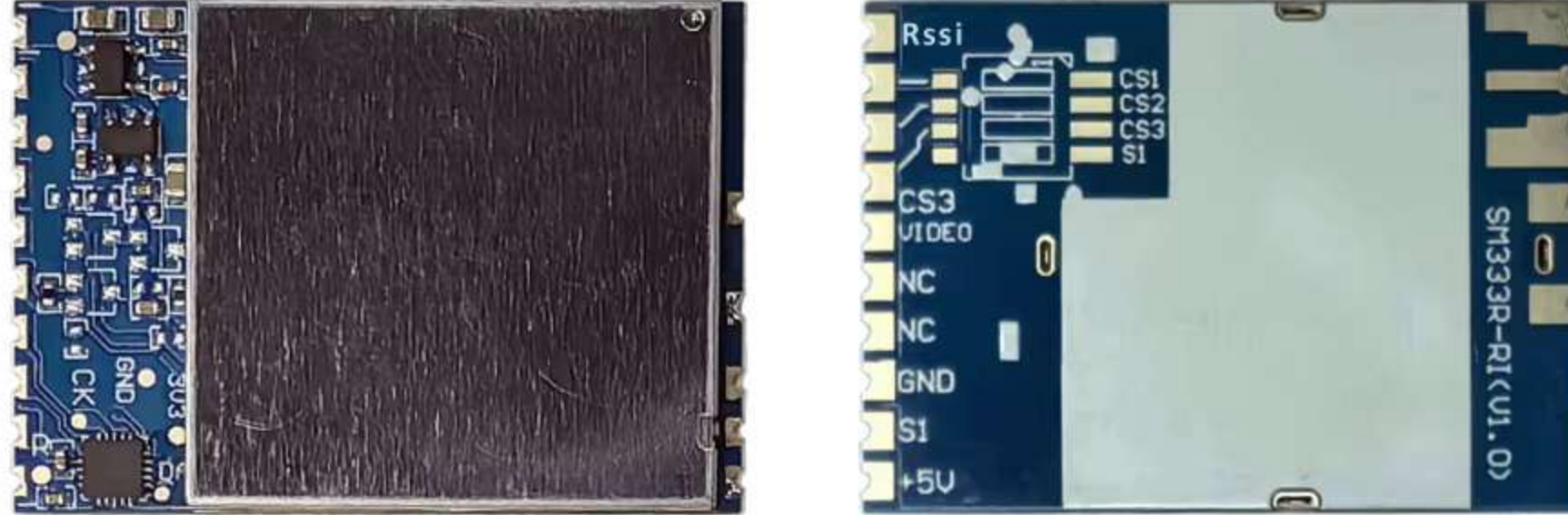


SM333R-RI FPV Receiver Mode (3.3G Toggle Switch mode)

SM333R-RI is a module that supports CVBS signal output, featuring low latency and strong anti-interference capabilities. It is widely used in fields such as FPV drone flight, outdoor entertainment, racing competitions, and professional aerial photography, the toggle switch control frequency。



1、General Specifications

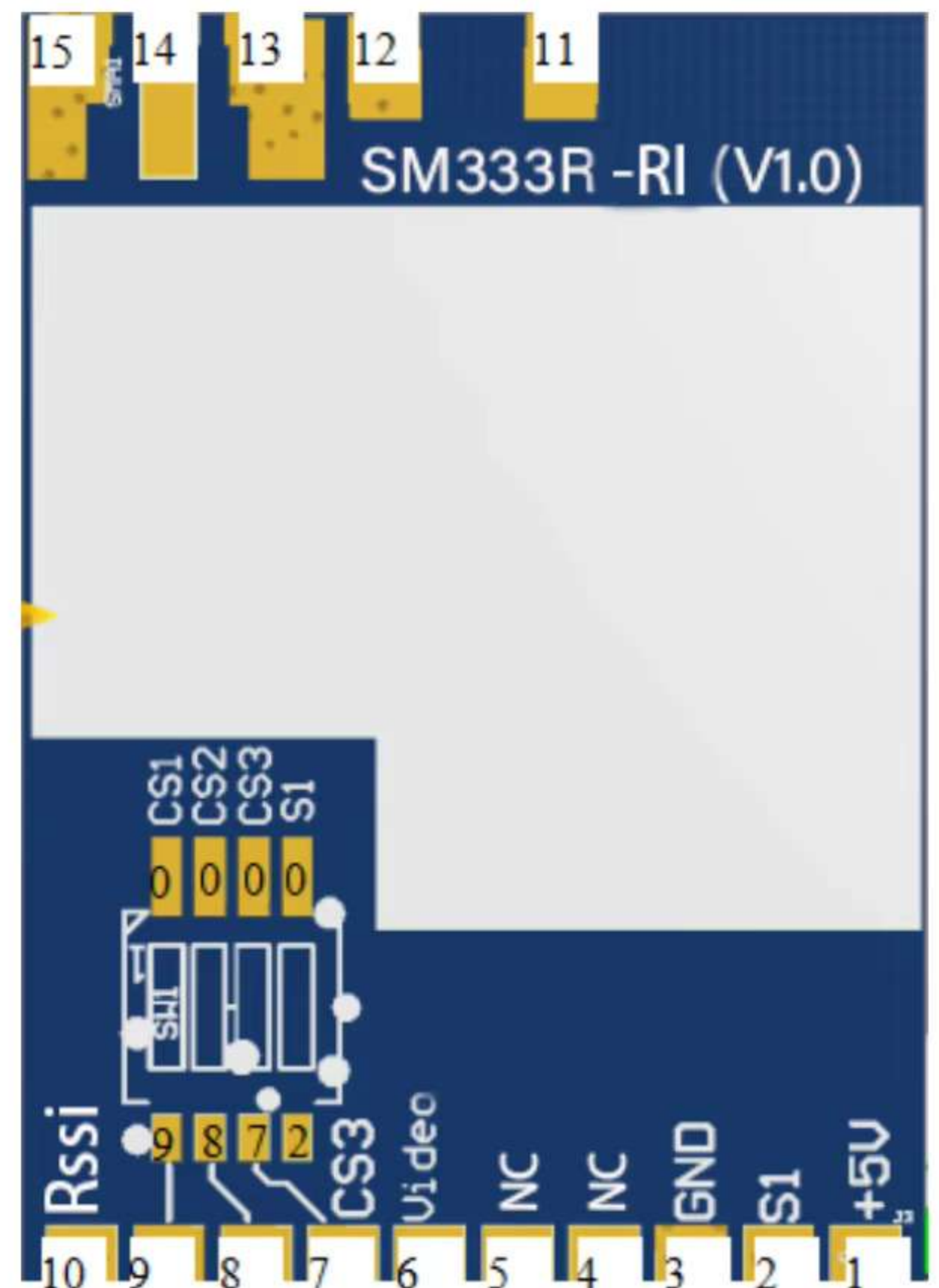
Item	SPEC.
DC Characteristics	
Input voltage	DC +5V
Consumption current	420mA±30mA @5V
Environmental Specification	
Operating Temperature	-10~+65 °C
Storing Temperature	-30~+85 °C
Operating humidity)	85%RH
RF Characteristics sunflytek.en.alibaba.com	
Receiving frequency range	3170~3470MHz
Demodulation system	FM / PLL
IF	480MHz
ANT. input impedance	50Ω, Typ.
LO Frequency stabilization	±200kHz
LO Frequency Precision	±200kHz
LO Control	PLL
Input LO Leak	-65dBm
Receiving Sensitivity	-95dBm ± 5dBm
Channel Customize	16
Channel Frequency	A: 3330/3350/3370/3390/3410/3430/3450/3470(MHz) B: 3170/3190/3210/3230/3250/3270/3290/3310(MHz)
Video Characteristics	
Video Output impedance	75Ω, Type
Video Output Level	1V±0.2Vp-p type
Video Frequency Response	±5 dB, Max. 50Hz ~ 6MHz
Differential Gain	±5 %, Max
Differential Phase	±5 Deg., Max
3dBm IF WIDEBAND	16.5MHz
S/N	38dB Min
RSSI	22mv/dBm
ANT. input impedance	50Ω, Type
Dimension	37mm(L)*26.3mm(W)*4.8mm(H)
Weight	Approx.: 5.2g

2、Function introduce

1) Pin function

Pin NO.	Function Description
1	+5V: power supply DC 5V
2	S1: Frequency SET (refer to Channel Table)
3	GND
4	NC
5	NC
6	Video: CVBS video output
7	CS3: Frequency SET (refer to Channel Table)
8	CS2: Frequency SET (refer to Channel Table)
9	CS1: Frequency SET (refer to Channel Table)
10	RSSI: RF signal strength
11	GND
12	GND
13	GND
14	Antenna signal input port
15	GND

PAD NO.	Function Description
2	Connected together with S1 pad
7	Connected together with CS3 pad
8	Connected together with CS2 pad
9	Connected together with CS1 pad
0	Connected together with GND



2) Channel Table (Low level: 0, High level: 1)

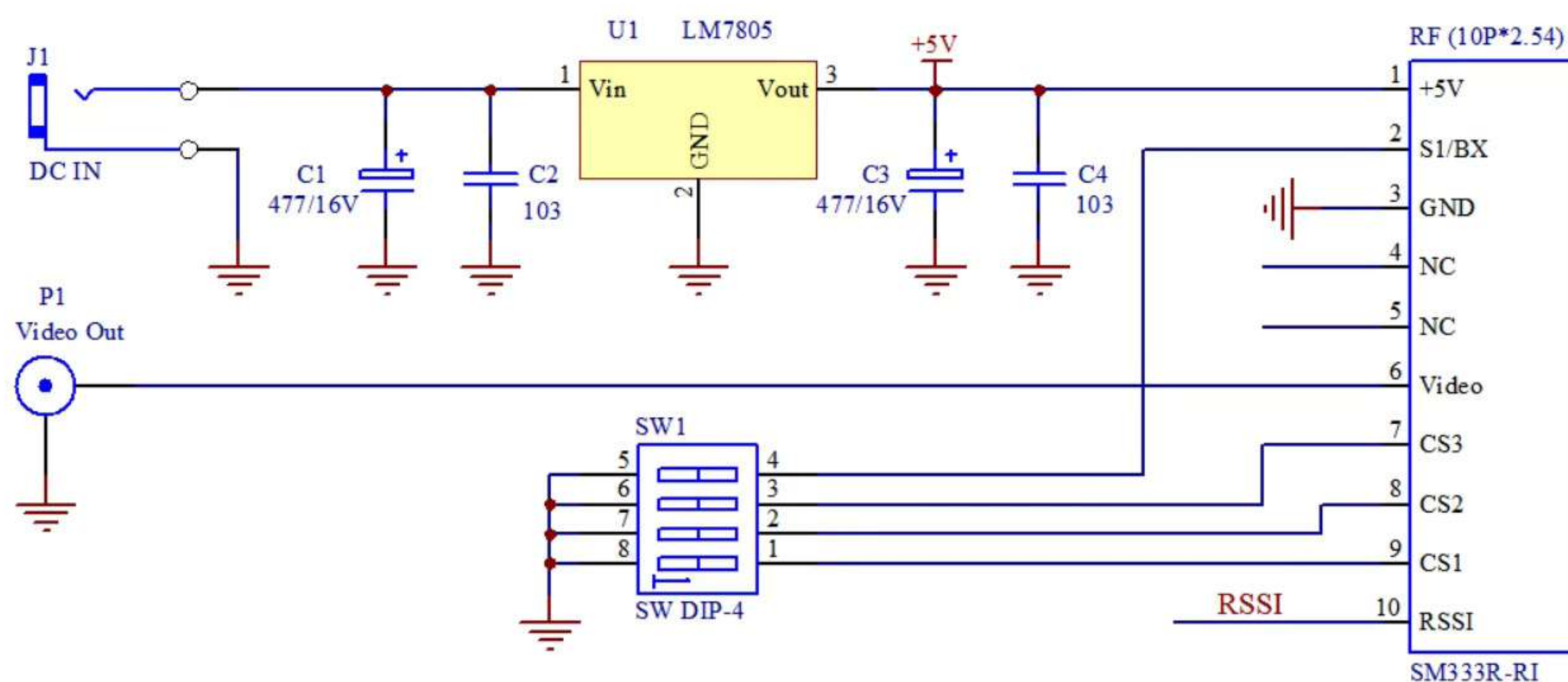
CHANNEL / LO	CS3	CS2	CS1	S1	Frequency group
A1: 3330MHz	0	0	0	1	A (S1=H)
A2: 3350MHz	0	0	1	1	
A3: 3370MHz	0	1	0	1	
A4: 3390MHz	0	1	1	1	
A5: 3410MHz	1	0	0	1	
A6: 3430MHz	1	0	1	1	
A7: 3450MHz	1	1	0	1	
A8: 3470MHz	1	1	1	1	
B1: 3170MHz	0	0	0	0	B (S1=L)
B2: 3190MHz	0	0	1	0	
B3: 3210MHz	0	1	0	0	
B4: 3230MHz	0	1	1	0	
B5: 3250MHz	1	0	0	0	
B6: 3270MHz	1	0	1	0	
B7: 3290MHz	1	1	0	0	
B8: 3310MHz	1	1	1	0	

3) RSSI output voltage (for reference only)

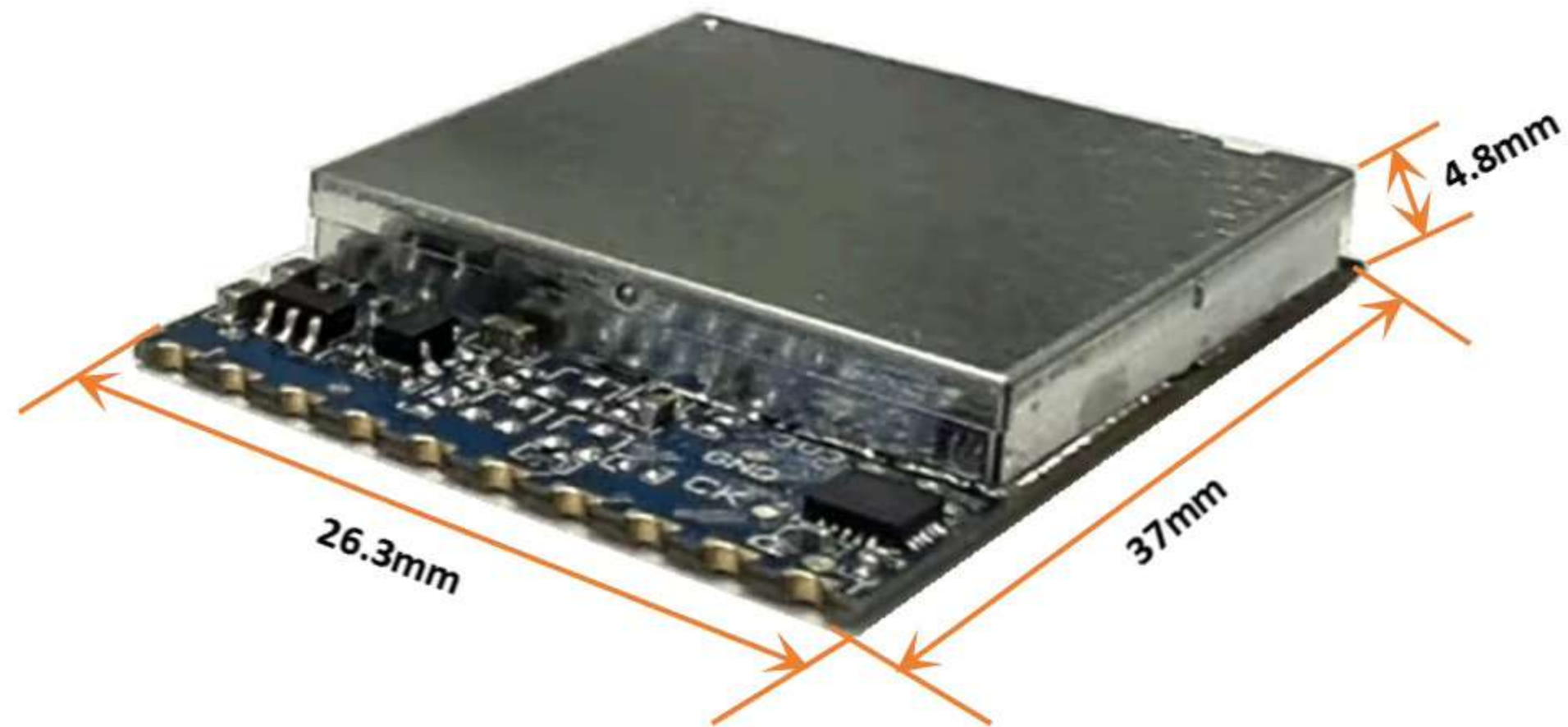
Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage
3330MHz	-136dBm	0.75V	3350MHz	-136dBm	0.75V	3370MHz	-136dBm	0.75V	3390MHz	-136dBm	0.77V
	-95dBm	0.92V		-95dBm	0.92V		-95dBm	0.92V		-95dBm	0.95V
	-90dBm	1.14V		-90dBm	1.13V		-90dBm	1.14V		-90dBm	1.19V
	-85dBm	1.69V		-85dBm	1.41V		-85dBm	1.42V		-85dBm	1.47V
	-80dBm	1.97V		-80dBm	1.68V		-80dBm	1.69V		-80dBm	1.76V
	-75dBm	2.25V		-75dBm	1.97V		-75dBm	1.97V		-75dBm	2.03V
	-70dBm	2.53V		-70dBm	2.25V		-70dBm	2.25V		-70dBm	2.31V
	-65dBm	2.64V		-65dBm	2.53V		-65dBm	2.54V		-65dBm	2.59V
	-60dBm	2.76V		-60dBm	2.78V		-60dBm	2.65V		-60dBm	2.76V
	-55dBm	2.77V		-55dBm	2.78V		-55dBm	2.78V		-55dBm	2.78V
	-50dBm	2.77V		-50dBm	2.79V		-50dBm	2.78V		-50dBm	2.78V
	-45dBm	2.77V		-45dBm	2.79V		-45dBm	2.79V		-45dBm	2.79V
	-40dBm	2.77V		-40dBm	2.79V		-40dBm	2.79V		-40dBm	2.79V
Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage
3410MHz	-136dBm	0.79V	3430MHz	-136dBm	0.84V	3450MHz	-136dBm	0.84V	3470MHz	-136dBm	0.87V
	-95dBm	0.96V		-95dBm	1.04V		-95dBm	1.05V		-95dBm	1.09V
	-90dBm	1.19V		-90dBm	1.26V		-90dBm	1.28V		-90dBm	1.33V
	-85dBm	1.47V		-85dBm	1.54V		-85dBm	1.56V		-85dBm	1.62V
	-80dBm	1.76V		-80dBm	1.82V		-80dBm	1.83V		-80dBm	1.89V
	-75dBm	2.03V		-75dBm	2.1V		-75dBm	2.12V		-75dBm	2.17V
	-70dBm	2.31V		-70dBm	2.38V		-70dBm	2.4V		-70dBm	2.45V
	-65dBm	2.59V		-65dBm	2.68V		-65dBm	2.6V		-65dBm	2.67V
	-60dBm	2.64V		-60dBm	2.76V		-60dBm	2.75		-60dBm	2.73V
	-55dBm	2.76V		-55dBm	2.77V		-55dBm	2.76V		-55dBm	2.78V
	-50dBm	2.78V		-50dBm	2.78V		-50dBm	2.78V		-50dBm	2.78V
	-45dBm	2.78V		-45dBm	2.78V		-45dBm	2.80V		-45dBm	2.80V
	-40dBm	2.79V		-40dBm	2.79V		-40dBm	2.80V		-40dBm	2.80V

Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage
3170MHz	-136dBm	0.66V	3190MHz	-136dBm	0.69V	3210MHz	-136dBm	0.66V	3230MHz	-136dBm	0.68V
	-95dBm	0.81V		-95dBm	0.83V		-95dBm	0.80V		-95dBm	0.83V
	-90dBm	1.01V		-90dBm	1.02V		-90dBm	1.01V		-90dBm	1.03V
	-85dBm	1.28V		-85dBm	1.30V		-85dBm	1.29V		-85dBm	1.30V
	-80dBm	1.56V		-80dBm	1.57V		-80dBm	1.56V		-80dBm	1.58V
	-75dBm	1.84V		-75dBm	1.86V		-75dBm	1.85V		-75dBm	1.87V
	-70dBm	2.11V		-70dBm	2.13V		-70dBm	2.13V		-70dBm	2.14V
	-65dBm	2.40V		-65dBm	2.41V		-65dBm	2.42V		-65dBm	2.43V
	-60dBm	2.68V		-60dBm	2.69V		-60dBm	2.70V		-60dBm	2.65V
	-55dBm	2.81V		-55dBm	2.78V		-55dBm	2.76V		-55dBm	2.76V
	-50dBm	2.81V		-50dBm	2.78V		-50dBm	2.76V		-50dBm	2.76V
	-45dBm	2.83V		-45dBm	2.81V		-45dBm	2.78V		-45dBm	2.78V
	-40dBm	2.83V		-40dBm	2.81V		-40dBm	2.78V		-40dBm	2.78V
Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage	Channel	Signal strength	RSSI Output voltage
3250MHz	-136dBm	0.67V	3270MHz	-136dBm	0.69V	3290MHz	-136dBm	0.7V	3310MHz	-136dBm	0.71V
	-95dBm	0.82V		-95dBm	0.86V		-95dBm	0.88V		-95dBm	0.90V
	-90dBm	1.02V		-90dBm	1.08V		-90dBm	1.09V		-90dBm	1.11V
	-85dBm	1.31V		-85dBm	1.36V		-85dBm	1.37V		-85dBm	1.39V
	-80dBm	1.58V		-80dBm	1.63V		-80dBm	1.64V		-80dBm	1.67V
	-75dBm	1.88V		-75dBm	1.91V		-75dBm	1.92V		-75dBm	1.95V
	-70dBm	2.16V		-70dBm	2.19V		-70dBm	2.20V		-70dBm	2.23V
	-65dBm	2.44V		-65dBm	2.48V		-65dBm	2.50V		-65dBm	2.51V
	-60dBm	2.65V		-60dBm	2.66V		-60dBm	2.66V		-60dBm	2.66V
	-55dBm	2.76V		-55dBm	2.76V		-55dBm	2.76V		-55dBm	2.76V
	-50dBm	2.76V		-50dBm	2.76V		-50dBm	2.76V		-50dBm	2.76V
	-45dBm	2.78V		-45dBm	2.78V		-45dBm	2.78V		-45dBm	2.78V
	-40dBm	2.78V		-40dBm	2.78V		-40dBm	2.78V		-40dBm	2.78V

3、Application Circuit



4、Dimension



5、Precautions for use

1. Before connecting the power supply, ensure that the correct voltage range and positive and negative poles are correct to avoid burning out the components.
2. Try to use antennas with good standing wave ratio and gain to achieve longer transmission distances.